Arithmetic Operations

Signed Integer

- 3 major representations:

  Sign and magnitude

  One’s complement

  Two’s complement
- Assumptions:

  4-bit machine word

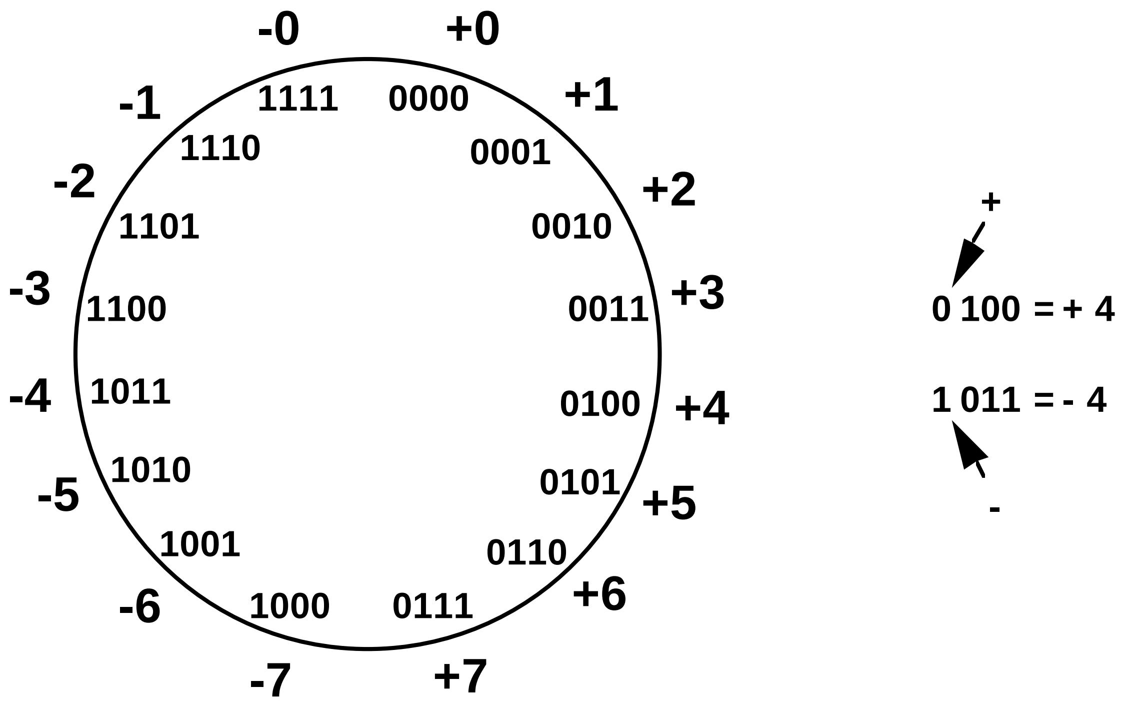
  16 different values can be represented

  Roughly half are positive, half are negative

Sign and Magnitude Representation

High order bit is sign: 0 = positive (or zero), 1 = negative
Three low order bits is the magnitude: 0 (000) thru 7 (111)
Number range for n bits = +/-2^{n-1} -1
Two representations for 0

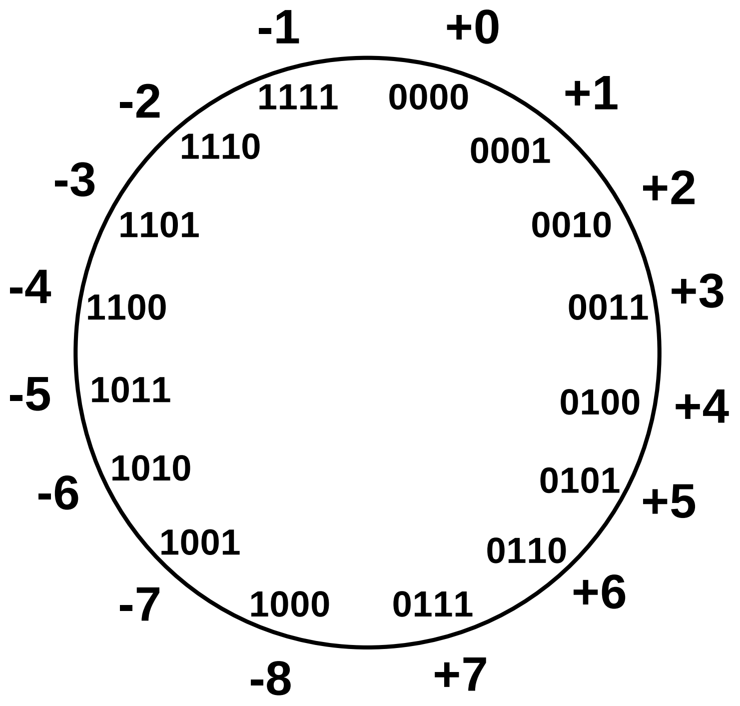
One’s Complement Representation

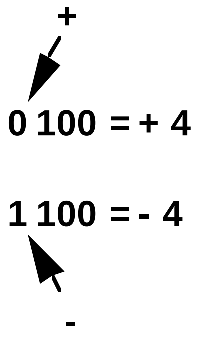


- Subtraction implemented by addition & 1's complement
- Still two representations of 0! This causes some problems
- Some complexities in addition

Two’s Complement Representation

like 1's comp except shifted one position clockwise





- Only one representation for 0
- One more negative number than positive number

Binary, Signed-Integer Representations

B	Values represented		
$b_3 b_2 b_1 b_0$	Sign and magnitude	1's complement	2's complement
0 1 1 1	+ 7	+ 7	+ 7
0 1 1 0	+ 6	+ 6	+ 6
0 1 0 1	+ 5	+ 5	+ 5
0 1 0 0	+ 4	+ 4	+ 4
0 0 1 1	+ 3	+ 3	+ 3
0 0 1 0	+ 2	+ 2	+ 2
0 0 0 1	+ 1	+ 1	+ 1
0 0 0 0	+ 0	+ 0	+ 0
1 0 0 0	- 0	- 7	- 8
1 0 0 1	- 1	- 6	- 7
1 0 1 0	- 2	- 5	- 6
1 0 1 1	- 3	- 4	- 5
1 1 0 0	- 4	- 3	- 4
1 1 0 1	- 5	- 2	- 3
1 1 1 0	- 6	- 1	- 2
1 1 1 1	- 7	- 0	- 1

Figure 2.1. Binary, signed-integer representations.

Addition and Subtraction – 2's Complement

If carry-in to the high order bit = carry-out then ignore carry

if carry-in differs from carry-out then overflow

4	0100	-4	1100
+ 3	0011	+ (-3)	1101
7	0111	-7	11001
4	0100	-4	1100
- 3	1101	+ 3	0011
1	10001	-1	1111

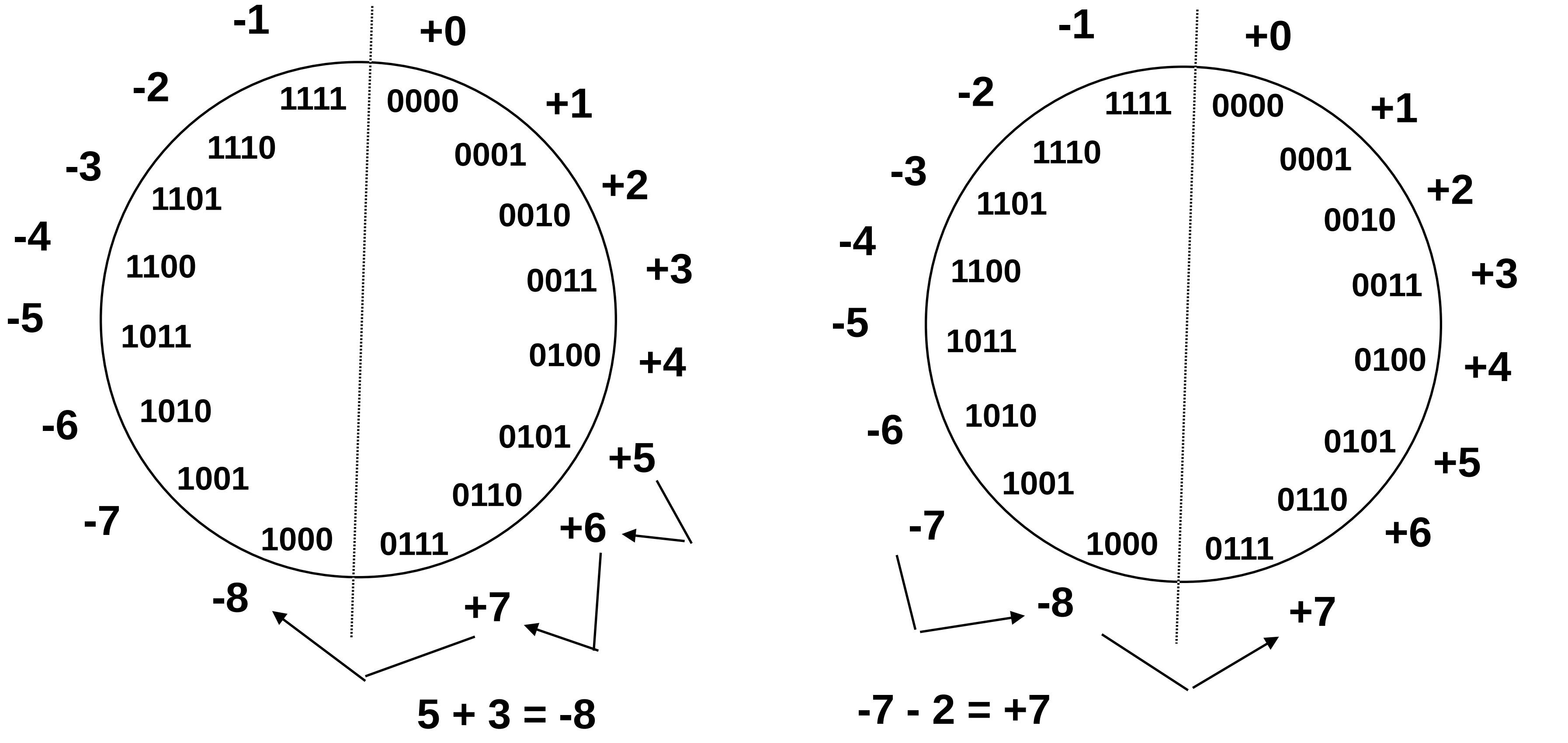
Simpler addition scheme makes two's complement the most common choice for integer number systems within digital systems

2’s-Complement Add and Subtract Operations

(a)	0 0 1 0	(+2)	(b)	0 1 0 0	(+4)
	+ 0 0 1 1	(+3)		+ 1 0 1 0	(- 6)
	0 1 0 1	(+5)		1 1 1 0	(- 2)
(c)	1 0 1 1	(- 5)	(d)	0 1 1 1	(+7)
	+ 1 1 1 0	(- 2)		+ 1 1 0 1	(- 3)
	1 0 0 1	(- 7)		0 1 0 0	(+4)
(e)	1 1 0 1	(- 3)		1 1 0 1	
	- 1 0 0 1	(- 7)	⇒	+ 0 1 1 1	
				0 1 0 0	(+4)
(f)	0 0 1 0	(+2)		0 0 1 0	
	- 0 1 0 0	(+ 4)	⇒	+ 1 1 0 0	
				1 1 1 0	(- 2)
(g)	0 1 1 0	(+6)		0 1 1 0	
	- 0 0 1 1	(+3)	⇒	+ 1 1 0 1	
				0 0 1 1	(+3)
(h)	1 0 0 1	(- 7)		1 0 0 1	
	- 1 0 1 1	(- 5)	⇒	+ 0 1 0 1	
				1 1 1 0	(- 2)
(i)	1 0 0 1	(- 7)		1 0 0 1	
	- 0 0 0 1	(+1)	⇒	+ 1 1 1 1	
				1 0 0 0	(- 8)
(j)	0 0 1 0	(+2)		0 0 1 0	
	- 1 1 0 1	(- 3)	⇒	+ 0 0 1 1	
				0 1 0 1	(+ 5)

Figure 2.4. 2's-complement Add and Subtract operations.

Overflow - Add two positive numbers to get a negative number or two negative numbers to get a positive number



Overflow Conditions

	0 1 1 1
5	0 1 0 1
3	0 0 1 1
-8	1 0 0 0

Overflow

	1 0 0 0
-7	1 0 0 1
-2	1 1 0 0
7	1 0 1 1 1

Overflow

	0 0 0 0
5	0 1 0 1
2	0 0 1 0
7	0 1 1 1

No overflow

	1 1 1 1
-3	1 1 0 1
-5	1 0 1 1
-8	1 1 0 0 0

No overflow

- **Overflows can occur when the sign of the two operands is the same.**
- **Overflow occurs if the sign of the result is different from the sign of the operands.**
- **Overflow when carry-in to the high-order bit does not equal carry out**

Addition and Subtraction of Signed Numbers

Addition/subtraction of signed Numbers

x_i	y_i	Carry-in c_i	Sum s_i	Carry-out c_{i+1}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$s_i = \bar{x}_i \bar{y}_i c_i + \bar{x}_i y_i \bar{c}_i + x_i \bar{y}_i \bar{c}_i + x_i y_i c_i = x_i \oplus y_i \oplus c_i$$

$$c_{i+1} = y_i c_i + x_i c_i + x_i y_i$$

At the i^{th} stage:

Input:

c_i is the carry-in

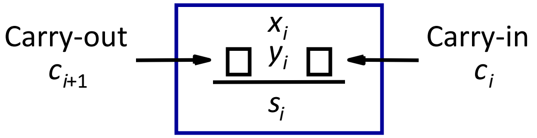
Output:

s_i is the sum

c_{i+1} carry-out to $(i+1)^{st}$ state

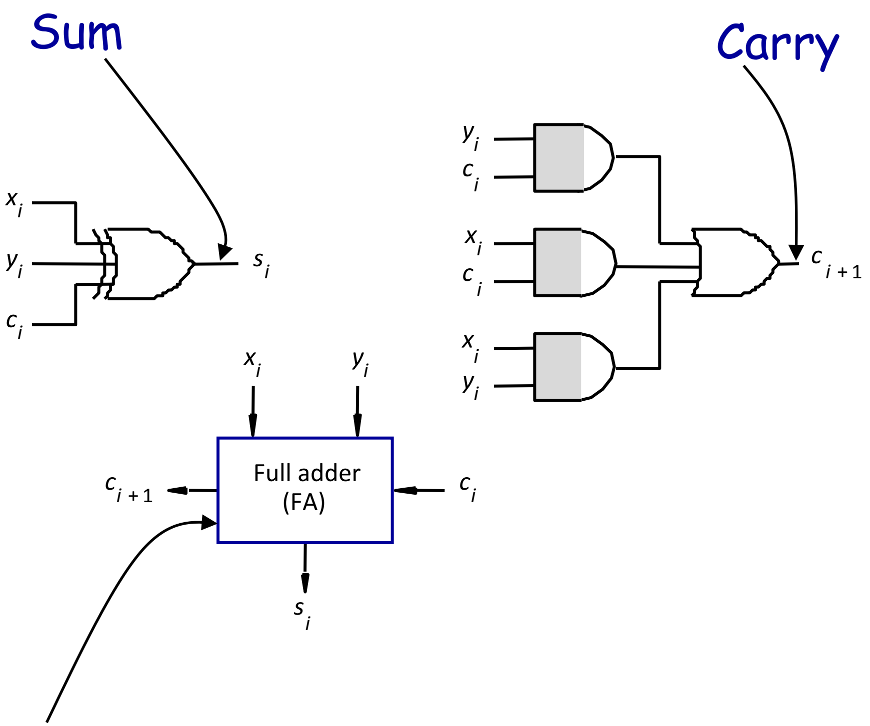
E xample:

$$\begin{array}{r} X \\ +\ Y \\ \hline Z \end{array} = \begin{array}{r} 7 \\ +\ 6 \\ \hline 13 \end{array} = \begin{array}{rrrr} 0 & 1 & \boxed{1} & 1 \\ +\ {}_0 0 & {}_1 1 & {}_1 1 & {}_0 0 \\ \hline 1 & 1 & 0 & 1 \end{array}$$



Legend for stage i

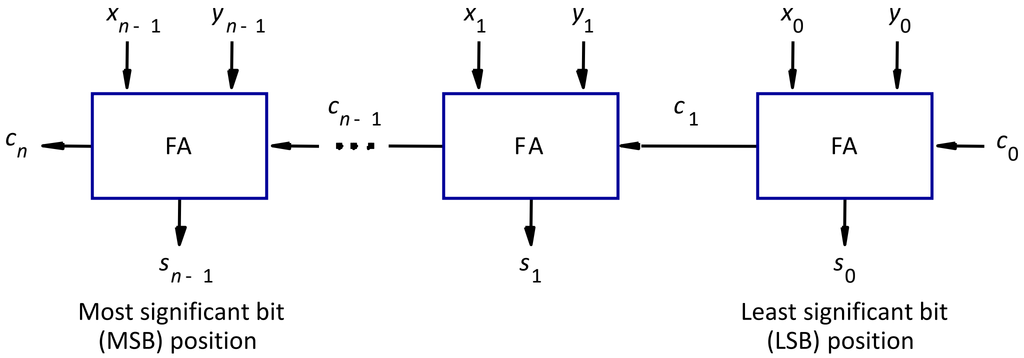
Addition logic for a single stage



Full Adder (FA): Symbol for the complete circuit
for a single stage of addition.

n-bit adder

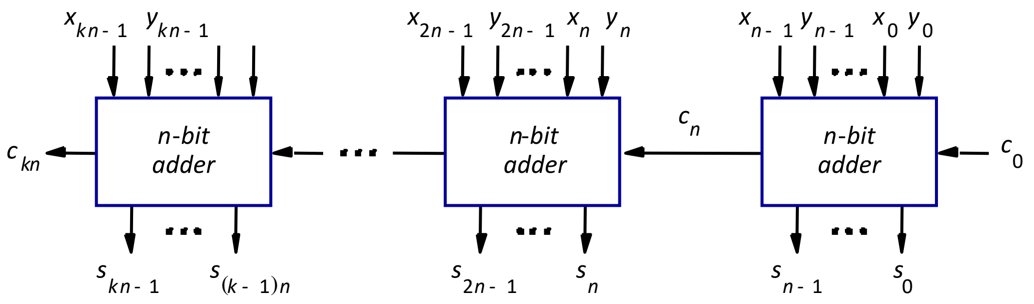
- Cascade n full adder (FA) blocks to form a n-bit adder.
- Carries propagate or ripple through this cascade, [n-bit ripple carry adder.]



Carry-in c_0 into the LSB position provides a convenient way to perform subtraction.

K n-bit adder

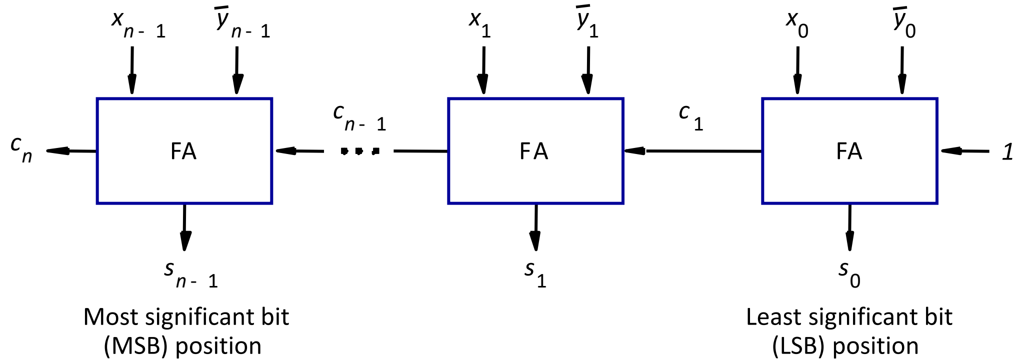
K n-bit numbers can be added by cascading k n-bit adders.



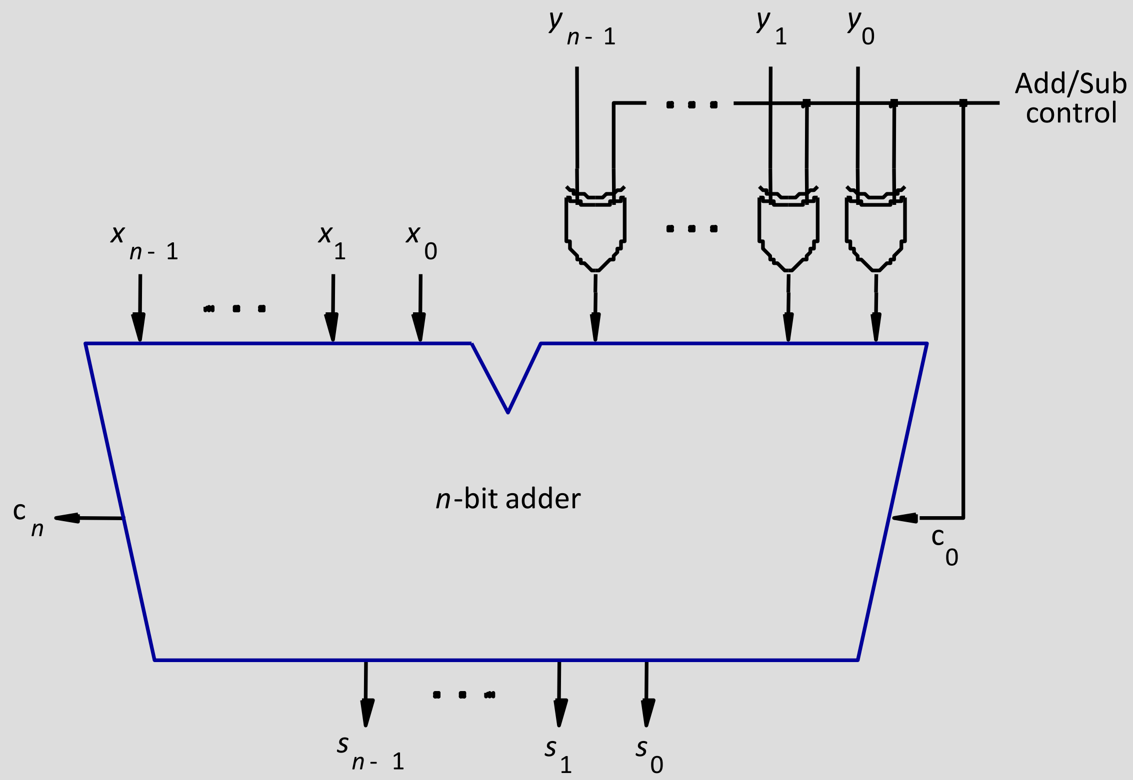
Each n-bit adder forms a block, so this is cascading of blocks.
Carries ripple or propagate through blocks, Blocked Ripple Carry Adder

n-bit subtractor

- Recall $X - Y$ is equivalent to adding 2's complement of Y to X.
- 2's complement is equivalent to 1's complement + 1.
- $\mathrm{X} - \mathrm{Y} = \mathrm{X} + \overline{\mathrm{Y}} + 1$
- 2's complement of positive and negative numbers is computed similarly.



n-bit adder/subtractor (contd..)



- Add/sub control = 0, addition.
- Add/sub control = 1, subtraction.

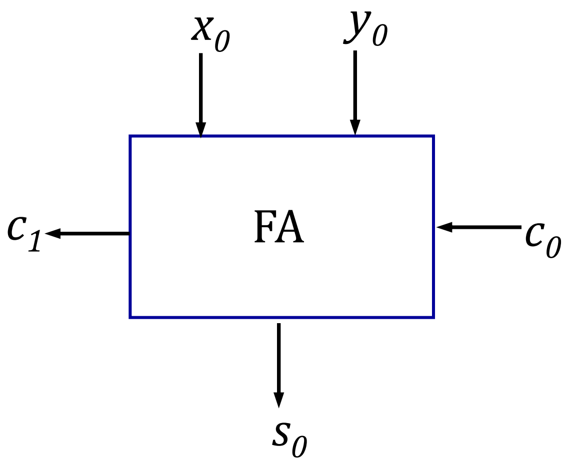
Detecting overflows

- Overflows can only occur when the sign of the two operands is the same.
- Overflow occurs if the sign of the result is different from the sign of the operands.
- Recall that the MSB represents the sign.
  - x_{n-1}, y_{n-1}, s_{n-1} represent the sign of operand *x,* operand *y* and result sum *s* respectively.
- Circuit to detect overflow can be implemented by the following logic expressions:

$$Overflow = x_{n-1} y_{n-1} \bar{s}_{n-1} + \bar{x}_{n-1} \bar{y}_{n-1} s_{n-1}$$

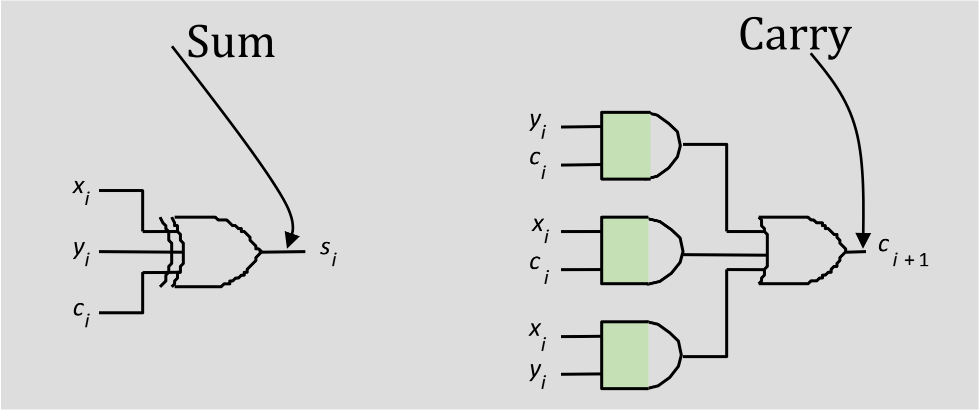
$$Overflow = c_n \oplus c_{n-1}$$

Computing the add time



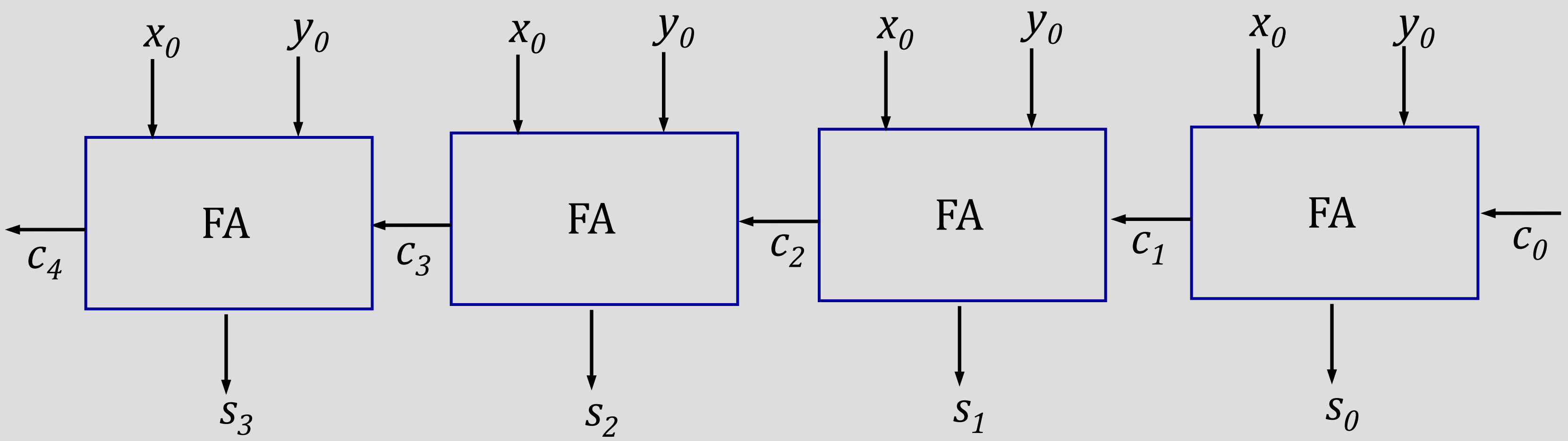
Consider 0^{th} stage:

- c_1 is available after 2 gate delays.
- s_1 is available after 1 gate delay.



Computing the add time (contd..)

Cascade of 4 Full Adders, or a 4-bit adder



- s_0 available after 1 gate delays, c_1 available after 2 gate delays.
- s_1 available after 3 gate delays, c_2 available after 4 gate delays.
- s_2 available after 5 gate delays, c_3 available after 6 gate delays.
- s_3 available after 7 gate delays, c_4 available after 8 gate delays.

For an n-bit adder, s_{n-1} is available after $2n-1$ gate delays
c_n is available after $2n$ gate delays.

Fast addition

Recall the equations:

$$s_i = x_i \oplus y_i \oplus c_i$$

$$c_{i+1} = x_i y_i + x_i c_i + y_i c_i$$

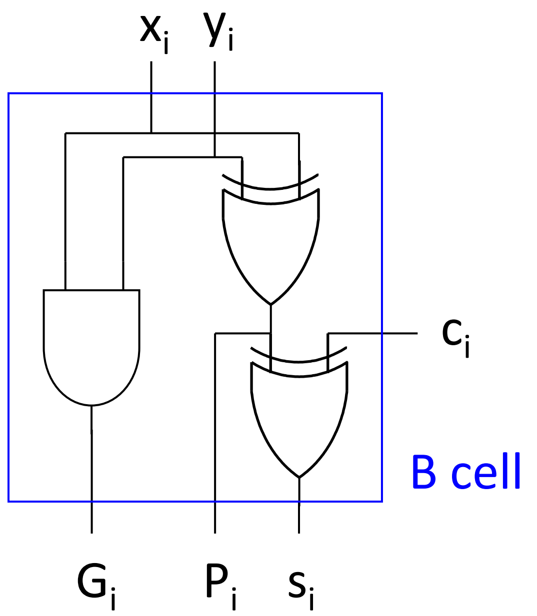
Second equation can be written as:

$$c_{i+1} = x_i y_i + (x_i + y_i) c_i$$

We can write:

$$c_{i+1} = G_i + P_i c_i$$

$$where\ G_i = x_i y_i\ and\ P_i = x_i + y_i$$



- G_i is called generate function and P_i is called propagate function
- G_i and P_i are computed only from x_i and y_i and not c_i, thus they can be computed in one gate delay after X and Y are applied to the inputs of an n-bit adder.
- Ci+1 = 1 only when Generate= 1 and Propagate = 0 or 1
- So we can modify the Pi

$$c_{i+1} = G_i + P_i c_i$$

$$where\ G_i = x_i y_i\ and\ P_i = x_i \oplus y_i$$

Carry lookahead

$$c_{i+1} = G_i + P_i c_i$$

$$c_i = G_{i-1} + P_{i-1} c_{i-1}$$

$$\Rightarrow c_{i+1} = G_i + P_i (G_{i-1} + P_{i-1} c_{i-1})$$

continuing

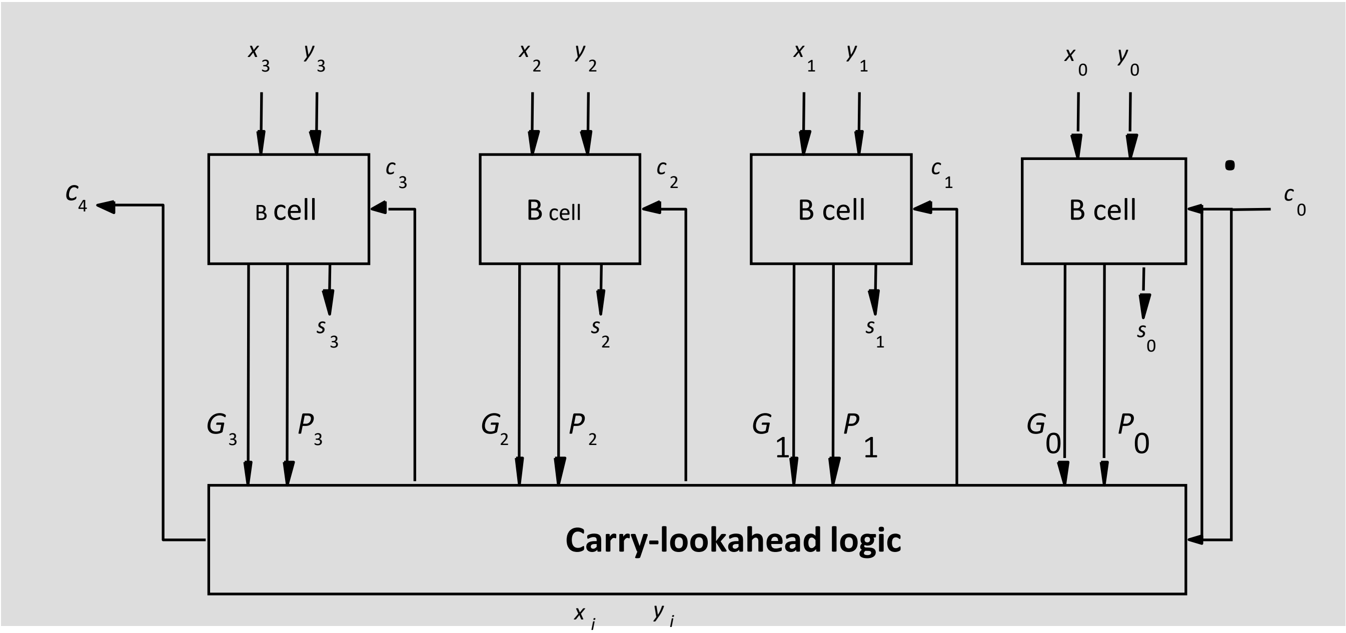
$$\Rightarrow c_{i+1} = G_i + P_i (G_{i-1} + P_{i-1} (G_{i-2} + P_{i-2} c_{i-2}))$$

until

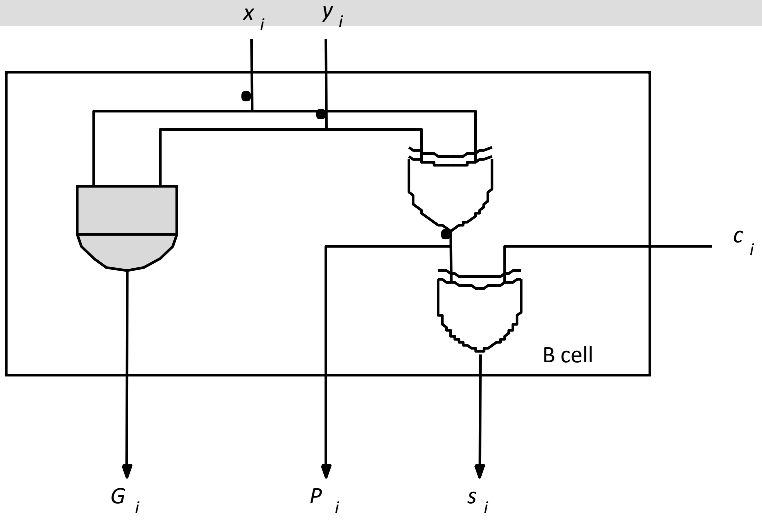
$$c_{i+1} = G_i + P_i G_{i-1} + P_i P_{i-1} G_{i-2} + .. + P_i P_{i-1} .. P_1 G_0 + P_i P_{i-1} \ldots P_0 c_0$$

- All carries can be obtained 3 gate delays after X, Y and c_0 are applied.
  - -One gate delay for P_i and G_i
  - -Two gate delays in the AND-OR circuit for c_{i+1}
- All sums can be obtained 1 gate delay after the carries are computed.
- Independent of n, n-bit addition requires only 4 gate delays.
- This is called Carry Lookahead adder.

Carry-lookahead adder



4-bit carry-lookahead adder



B-cell for a single stage

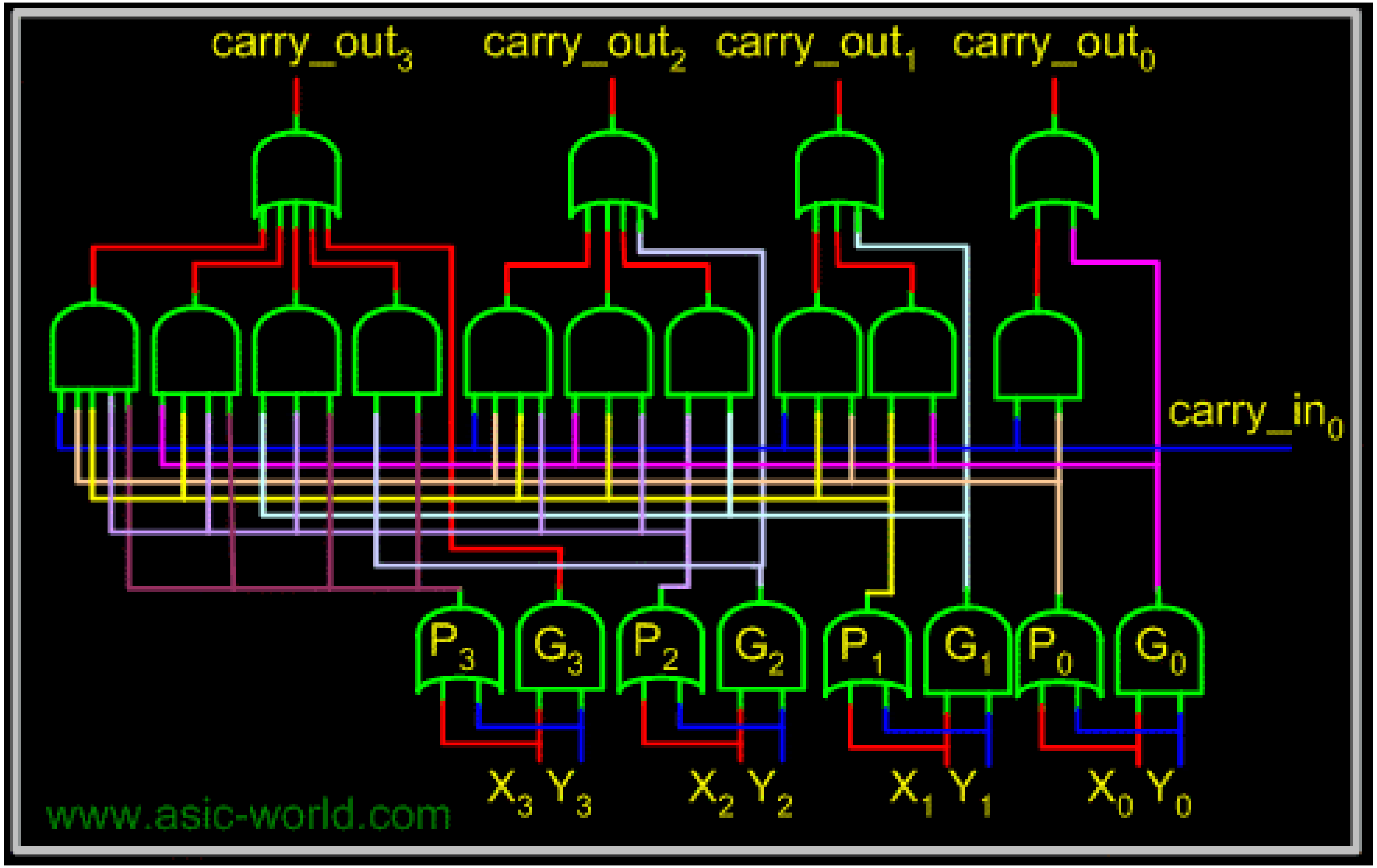
Carry lookahead adder (contd..)

- Performing n-bit addition in 4 gate delays independent of n is good only theoretically because of fan-in constraints.

$$c_{i+1} = G_i + P_i G_{i-1} + P_i P_{i-1} G_{i-2} + .. + P_i P_{i-1} .. P_1 G_0 + P_i P_{i-1} ... P_0 c_0$$

- Last AND gate and OR gate require a fan-in of (n+1) for a n-bit adder.
  - For a 4-bit adder ($n=4$) fan-in of 5 is required.
  - Practical limit for most gates.
- In order to add operands longer than 4 bits, we can cascade 4-bit Carry-Lookahead adders. Cascade of Carry-Lookahead adders is called Blocked Carry-Lookahead adder.
- Fan-in is the number of inputs a gate can handle.

4-bit carry-lookahead Adder



Blocked Carry-Lookahead adder

Carry-out from a 4-bit block can be given as:

$$c_4 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0c_0$$

Rewrite this as:

$$P_0^I = P_3P_2P_1P_0$$

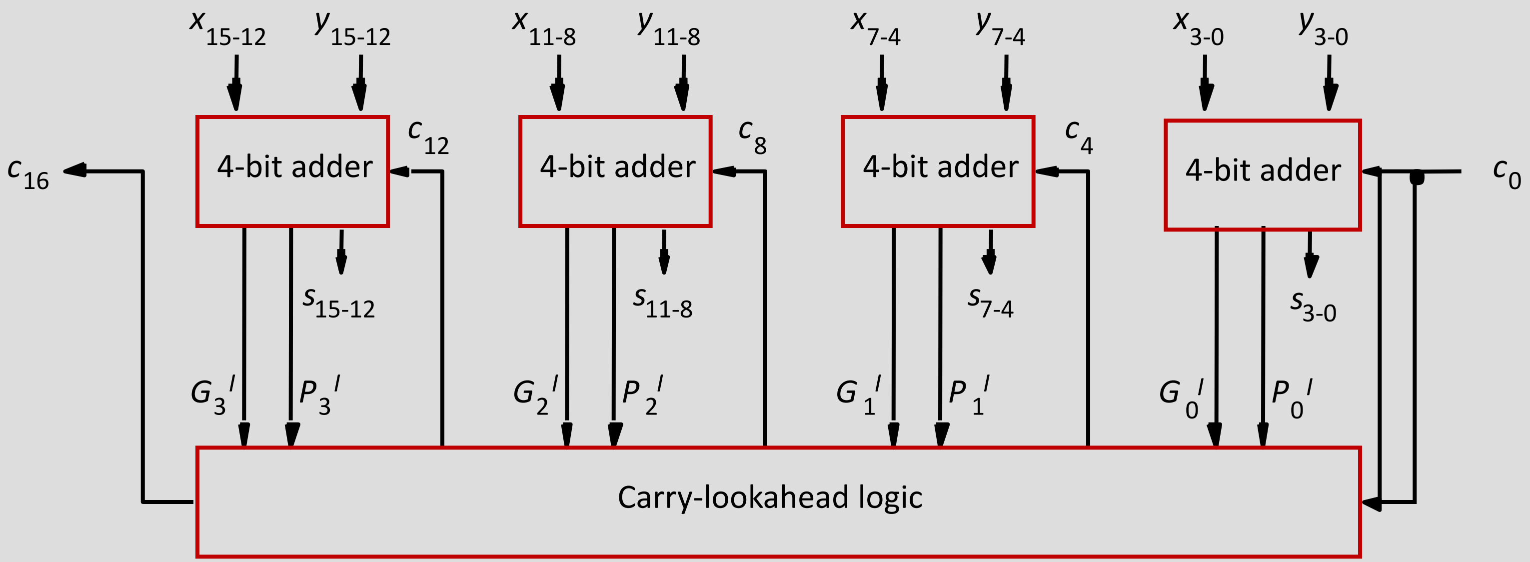
$$G_0^I = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0$$

Subscript I denotes the blocked carry lookahead and identifies the block.

Cascade 4 *4*-bit adders, c_{16} can be expressed as:

$$c_{16} = G_3^I + P_3^I G_2^I + P_3^I P_2^I G_1^I + P_3^I P_2^I P_1^0 G_0^I + P_3^I P_2^I P_1^0 P_0^0 c_0$$

Blocked Carry-Lookahead adder



After x_i, y_i and c_0 are applied as inputs:

- G_i and P_i for each stage are available after 1 gate delay.
- P^I is available after 2 and G^I after 3 gate delays.
- All carries are available after 5 gate delays.
- c_{16} is available after 5 gate delays.
- s_{15} which depends on c_{12} is available after 8 (5+3) gate delays (Recall that for a 4-bit carry lookahead adder, the last sum bit is available 3 gate delays after all inputs are available)

Multiplication

Multiplication of unsigned numbers

				1	1	0	1	(13) Multiplicand M
				1	0	1	1	(11) Multiplier Q
				1	1	0	1	
			1	1	0	1		
		0	0	0	0			
	1	1	0	1				
1	0	0	0	1	1	1	1	(143) Product P

Product of 2 *n*-bit numbers is at most a *2n*-bit number.

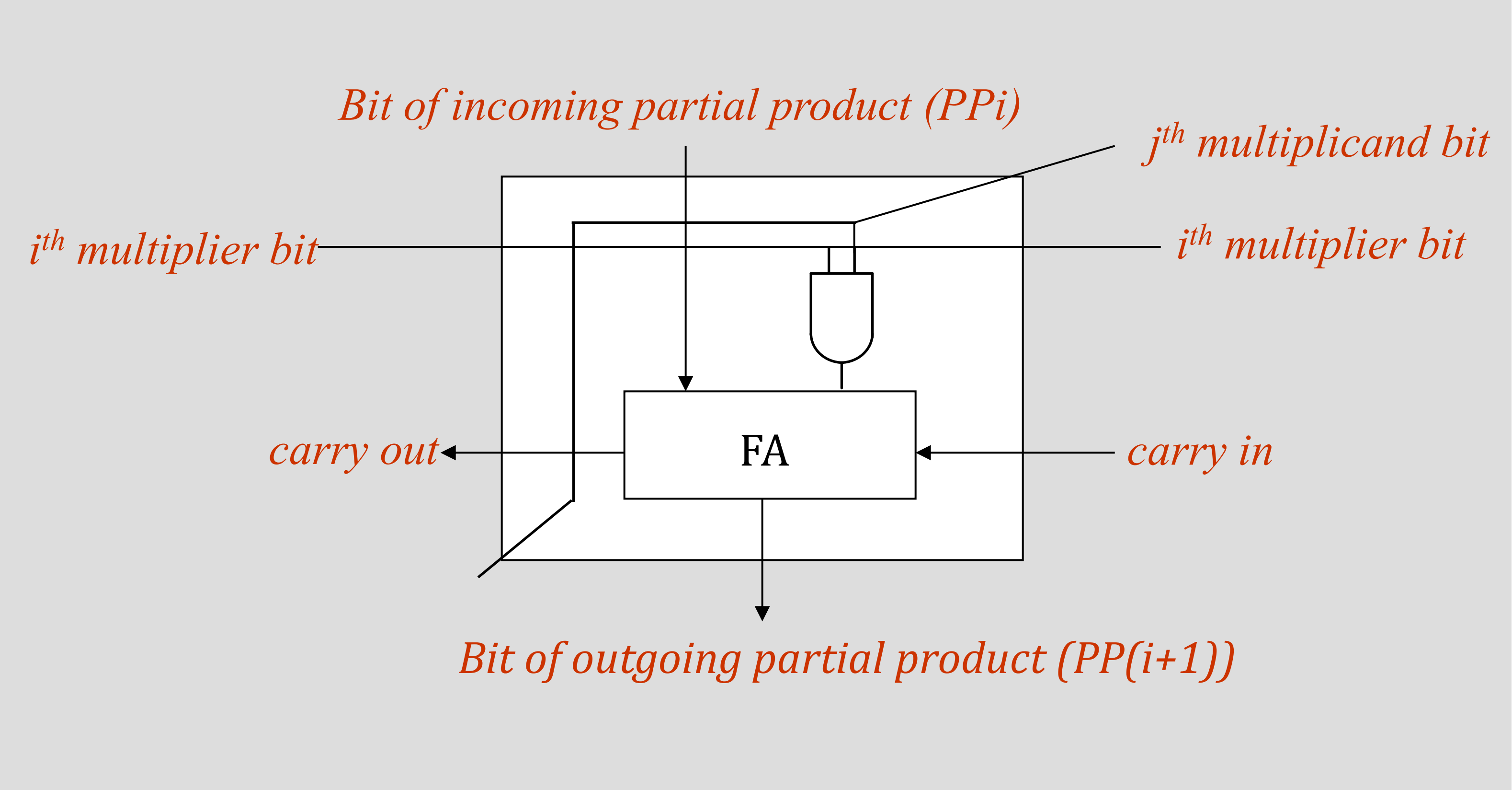
Unsigned multiplication can be viewed as addition of shifted versions of the multiplicand.

Multiplication of unsigned numbers (contd..)

- We added the partial products at end.
  - Alternative would be to add the partial products at each stage.
- Rules to implement multiplication are:
  - If the i^{th} bit of the multiplier is 1, shift the multiplicand and add the shifted multiplicand to the current value of the partial product.
  - Hand over the partial product to the next stage
  - Value of the partial product at the start stage is 0.

Multiplication of unsigned numbers

Typical multiplication cell

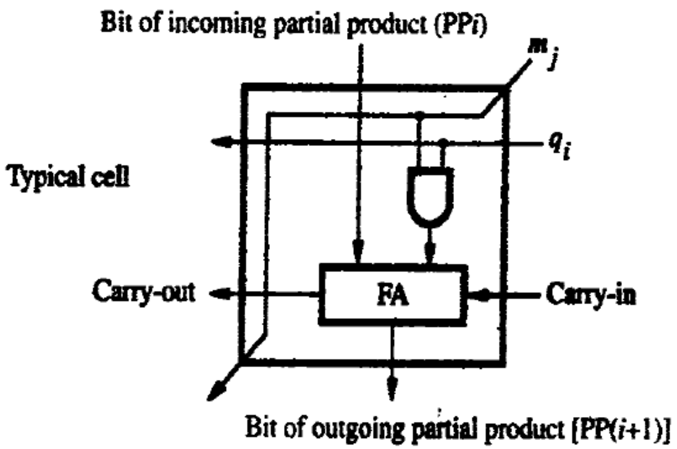


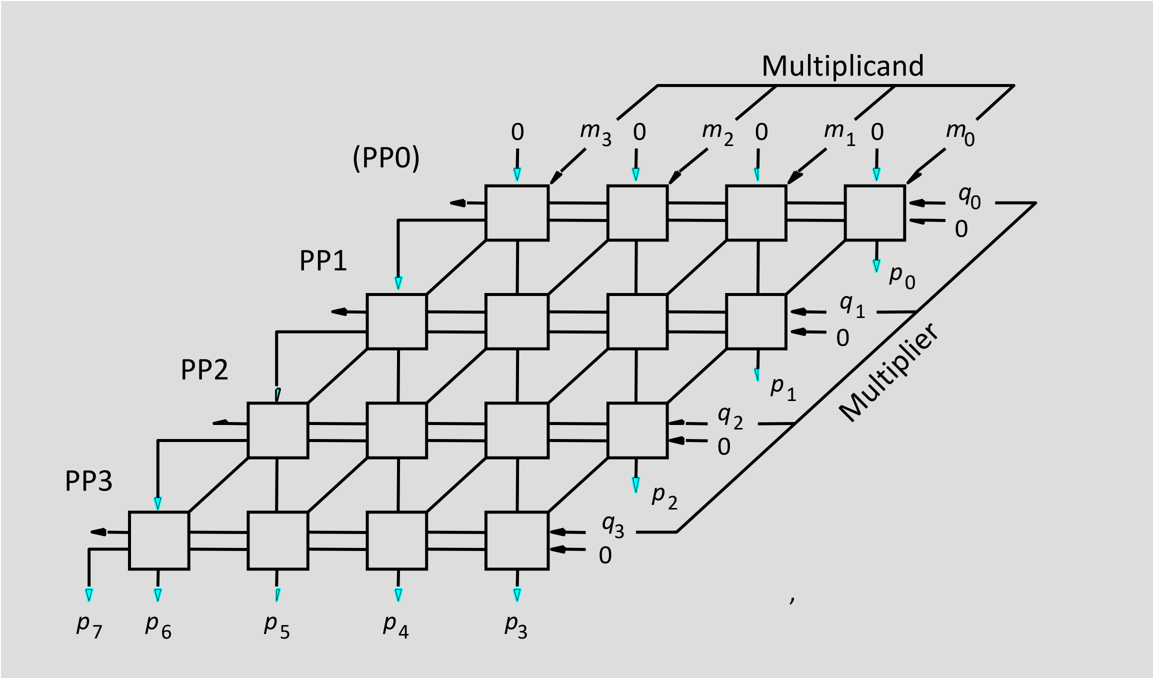
Typical multiplication cell

- The main component in each cell is a full adder, FA.
- The AND gate in each cell determines whether a multiplicand bit, mj , is added to the incoming partial-product bit, based on the value of the multiplier bit, qi.
- Each row i, where $0 \le i \le 3$, adds the multiplicand (appropriately shifted) to the incoming partial product, PPi, to generate the outgoing partial product, PP(i + 1), if $qi = 1$.
- If $qi = 0$, PPi is passed vertically downward unchanged.
- PP0 is all 0s, and PP4 is the desired product.
- The multiplicand is shifted left one position per row by the diagonal signal path.

Combinatorial array multiplier

Combinatorial array multiplier





Product is: $p_7, p_6, ..p_0$

Multiplicand is shifted by displacing it through an array of adders.

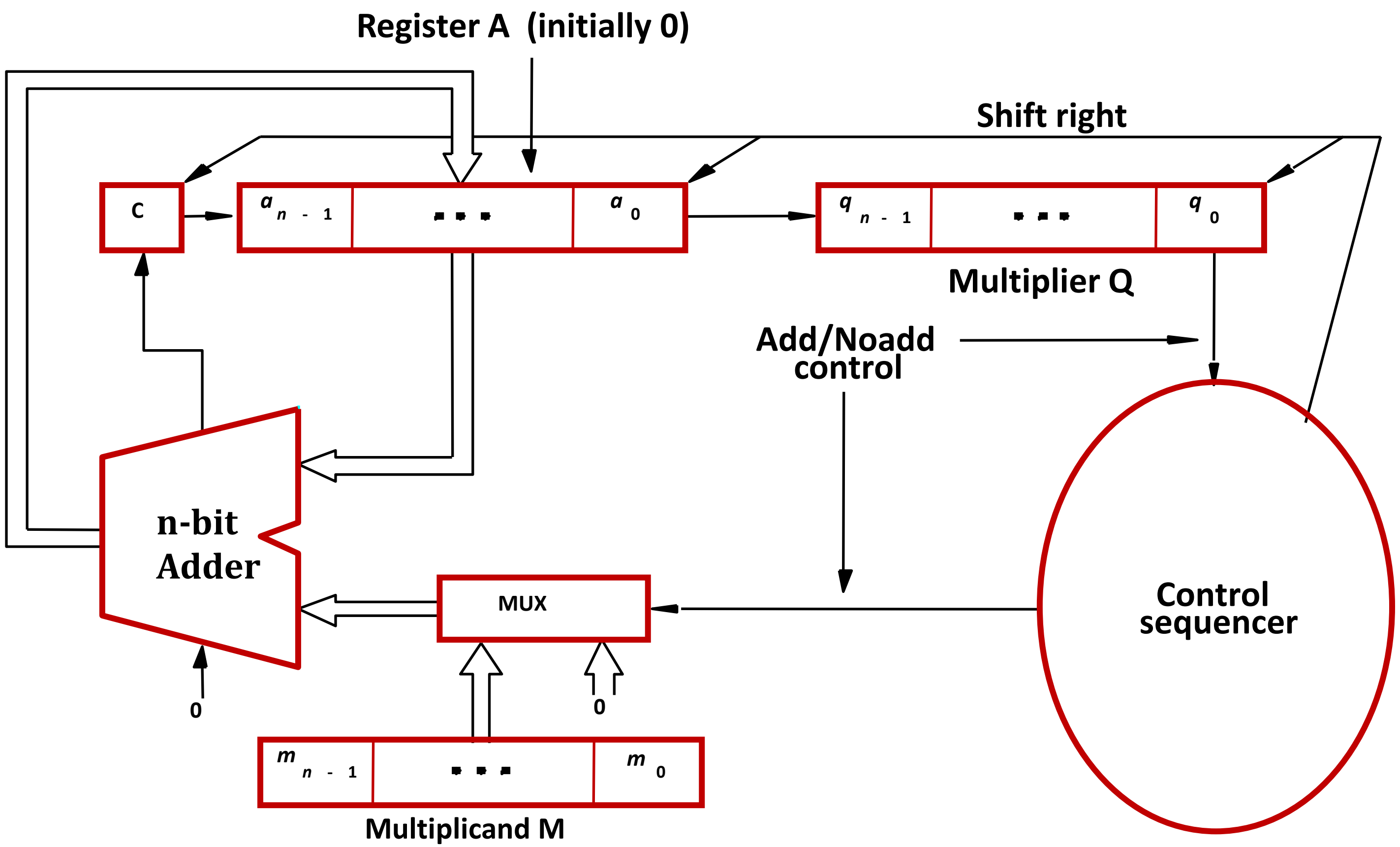
Combinatorial array multiplier (contd..)

- Combinatorial array multipliers are:
  - Extremely inefficient.
  - Have a high gate count for multiplying numbers of practical size such as 32-bit or 64-bit numbers.
  - Perform only one function, namely, unsigned integer product.
- Improve gate efficiency by using a mixture of combinatorial array techniques and sequential techniques requiring less combinational logic.

Sequential multiplication

- Recall the rule for generating partial products:
  - If the ith bit of the multiplier is 1, add the appropriately shifted multiplicand to the current partial product.
  - Multiplicand has been shifted <u>left</u> when added to the partial product.
- However, adding a left-shifted multiplicand to an unshifted partial product is equivalent to adding an unshifted multiplicand to a right-shifted partial product.

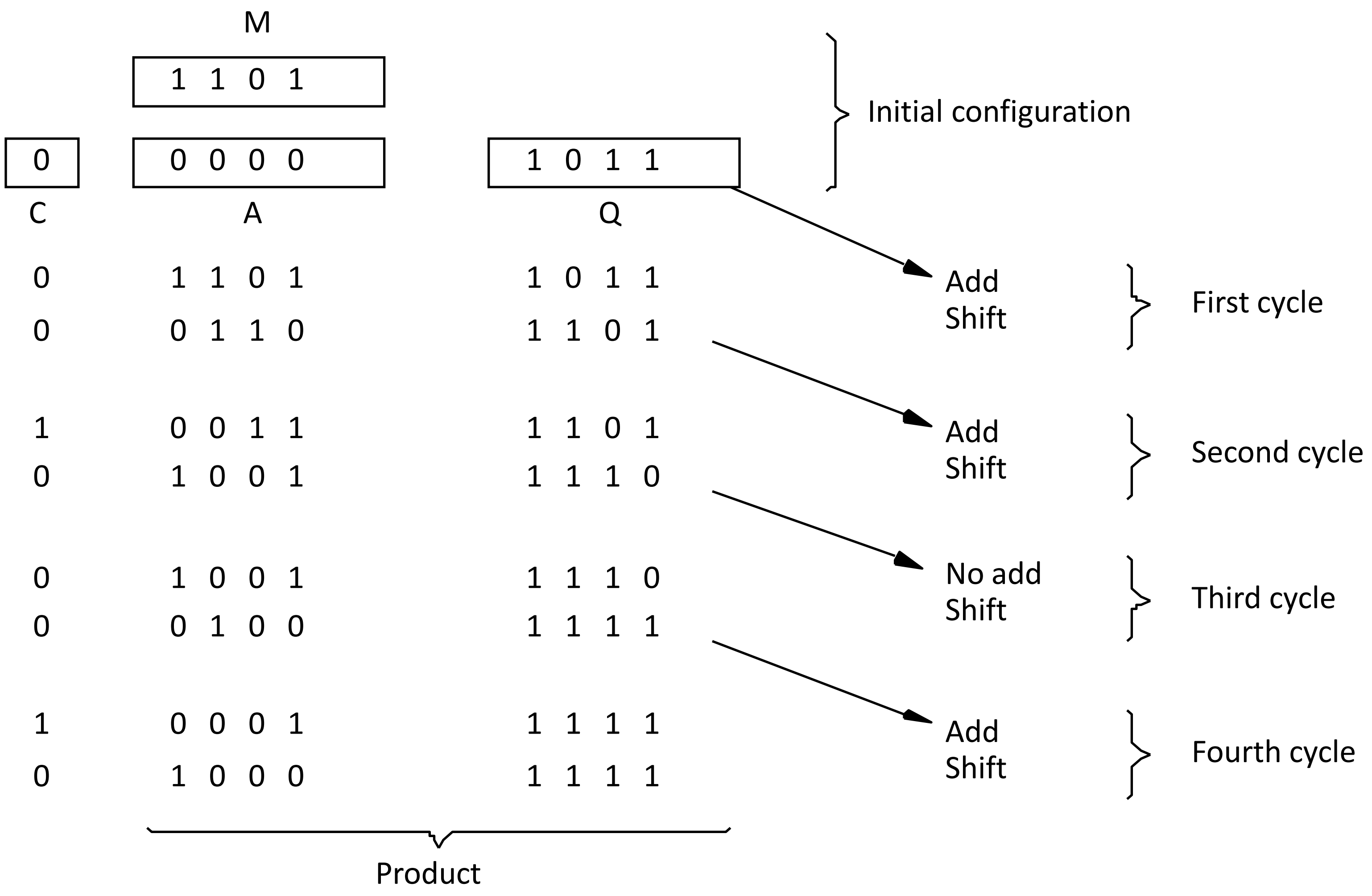
Sequential Circuit Multiplier



Sequential Circuit Multiplier

- Registers A and Q are shift registers, concatenated as shown. Together, they hold partial product PP*i* while multiplier bit *qi* generates the signal Add/No add.
- This signal causes the multiplexer MUX to select 0 when *qi* = 0, or to select the multiplicand M when *qi* = 1, to be added to PP*i* to generate PP(*i* + 1).
- The product is computed in *n* cycles. The partial product grows in length by one bit per cycle from the initial vector, PP0, of *n* 0s in register A.
- The carry-out from the adder is stored in flip-flop C, shown at the left end of register A.
- At the start, the multiplier is loaded into register Q, the multiplicand into register M, and C and A are cleared to 0. At the end of each cycle, C, A, and Q are shifted right one bit position to allow for growth of the partial product as the multiplier is shifted out of register Q.
- Because of this shifting, multiplier bit *qi* appears at the LSB position of Q to generate the Add/No add signal at the correct time, starting with *q*0 during the first cycle, *q*1 during the second cycle, and so on.
- After they are used, the multiplier bits are discarded by the right-shift operation. Note that the carry-out from the adder is the leftmost bit of PP(*i* + 1), and it must be held in the C flip-flop to be shifted right with the contents of A and Q.
- After *n* cycles, the high-order half of the product is held in register A and the low-order half is in register Q.

Sequential multiplication (contd..)



Signed Multiplication

Signed Multiplication

- Considering 2’s-complement signed operands, what will happen to (-13)×(+11) if following the same method of unsigned multiplication?

						1	0	0	1	1	(- 13)
						0	1	0	1	1	(+ 11)
	1	1	1	1	1	1	0	0	1	1	
	1	1	1	1	1	0	0	1	1		
Sign extension is shown in red	0	0	0	0	0	0	0	0			
	1	1	1	0	0	1	1				
	0	0	0	0	0	0					
	1	1	0	1	1	1	0	0	0	1	(- 143)

Sign extension of negative multiplicand.

Signed Multiplication

- For a negative multiplier, a straightforward solution is to form the 2's-complement of both the multiplier and the multiplicand and proceed as in the case of a positive multiplier.
- This is possible because complementation of both operands does not change the value or the sign of the product.
- A technique that works equally well for both negative and positive multipliers – Booth algorithm.

Booth Algorithm

- In general, in the Booth scheme, -1 times the shifted multiplicand is selected when moving from 0 to 1, and +1 times the shifted multiplicand is selected when moving from 1 to 0, as the multiplier is scanned from right to left.

0 0 1 0 1 1 0 0 1 1 1 0 1 0 1 1 0 0

0 +1 -1 +1 0 -1 0 +1 0 0 -1 +1 -1 +1 0 -1 0 0

Treats both positive and negative 2's-complement operands uniformly

Algorithm (scanning multiplier right to left)

0 to 1 => add −1 times shifted multiplicand

1 to 0 => add +1 times shifted multiplicand

0 to 0 or 1 to 1 => add 0

Booth Algorithm

```
 -4   1100
x 5   0101
---
-20
```

```
              1 1 0 0
            x 0 1 0 1
             +1-1+1-1
       ----------------
    0 0 0 0 0 1 0 0      2's complement of 1100 with signed extension 11111100
    1 1 1 1 1 0 0        Multiplicand 1100  with signed extension 11111100
    0 0 0 1 0 0          2's complement of 1100
+   1 1 1 0 0            Multiplicand 1100
  ---------------------
    1 1 1 0 1 1 0 0
    0 0 0 1 0 1 0 0  = -20
```

Booth Algorithm

```
   5     0101
 x-4     1100
 ---
 -20
```

```
          0 1 0 1
        x 1 1 0 0
          0-1 0 0
    ---------------
  0 0 0 0 0 0 0 0
  0 0 0 0 0 0 0
  1 1 1 0 1 1          2's complement of 0101
+ 0 0 0 0 0
-------------------
  1 1 1 0 1 1 0 0
  0 0 0 1 0 1 0 0 = -20
```

Booth Algorithm

- Consider in a multiplication, the multiplier is positive 0011110, how many appropriately shifted versions of the multiplicand are added in a standard procedure?

							0	1	0	1	1	0	1
							0	0	+1	+1	+1	+1	0
							0	0	0	0	0	0	0
						0	1	0	1	1	0	1	
					0	1	0	1	1	0	1		
				0	1	0	1	1	0	1			
			0	1	0	1	1	0	1				
		0	0	0	0	0	0	0					
	0	0	0	0	0	0	0						
0	0	0	1	0	1	0	1	0	0	0	1	1	0

Booth Algorithm

- Since 0011110 = 0100000 – 0000010, if we use the expression to the right, what will happen?

							0	1	0	1	1	0	1	
							0	+1	0	0	0	-1	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	
1	1	1	1	1	1	1	0	1	0	0	1	1		2's complement of the multiplicand
0	0	0	0	0	0	0	0	0	0	0	0			
0	0	0	0	0	0	0	0	0	0	0				
0	0	0	0	0	0	0	0	0	0					
0	0	0	1	0	1	1	0	1						
0	0	0	0	0	0	0	0							
0	0	0	1	0	1	0	1	0	0	0	1	1	0	

Booth Algorithm

	0	1	1	0	1	(+13)
X	1	1	0	1	0	(- 6)

$\Rightarrow$

					0	1	1	0	1	
					0	-1	+1	-1	0	
0	0	0	0	0	0	0	0	0	0	
1	1	1	1	1	0	0	1	1		
0	0	0	0	1	1	0	1			
1	1	1	0	0	1	1				
0	0	0	0	0	0					
1	1	1	0	1	1	0	0	1	0	(- 78)

Booth multiplication with a negative multiplier.

Booth Algorithm

Multiplier		Version of multiplicand selected by bit i
Bit i	Bit i-1	
0	0	$0 \times M$
0	1	$+1 \times M$
1	0	$-1 \times M$
1	1	$0 \times M$

Booth multiplier recoding table.

Booth Algorithm

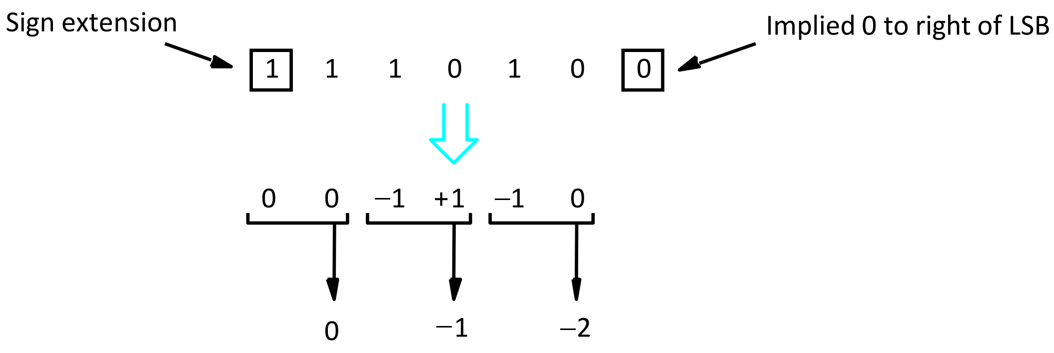
- Best case – a long string of 1's (skipping over 1s)
- Worst case – 0's and 1's are alternating

Worst-case multiplier	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1	+1	-1
Ordinary multiplier	1	1	0	0	0	1	0	1	1	0	1	1	1	1	0	0
	0	-1	0	0	+1	-1	+1	0	-1	+1	0	0	0	-1	0	0
Good multiplier	0	0	0	0	1	1	1	1	1	0	0	0	0	1	1	1
	0	0	0	+1	0	0	0	0	-1	0	0	0	+1	0	0	-1

Fast Multiplication

Bit-Pair Recoding of Multipliers

- Bit-pair recoding halves the maximum number of summands (versions of the multiplicand).



(a) Example of bit-pair recoding derived from Booth recoding

Bit-Pair Recoding of Multipliers

Multiplier bit-pair		Multiplier bit on the right	Multiplicand selected at position i
$i+1$	i	$i-1$	
0	0	0	0 X M
0	0	1	+ 1 X M
0	1	0	+ 1 X M
0	1	1	+ 2 X M
1	0	0	− 2 X M
1	0	1	− 1 X M
1	1	0	− 1 X M
1	1	1	0 X M

(b) Table of multiplicand selection decisions

Bit-Pair Recoding of Multipliers

```
    0 1 1 0 1   (+13)
×   1 1 0 1 0   (- 6)
```

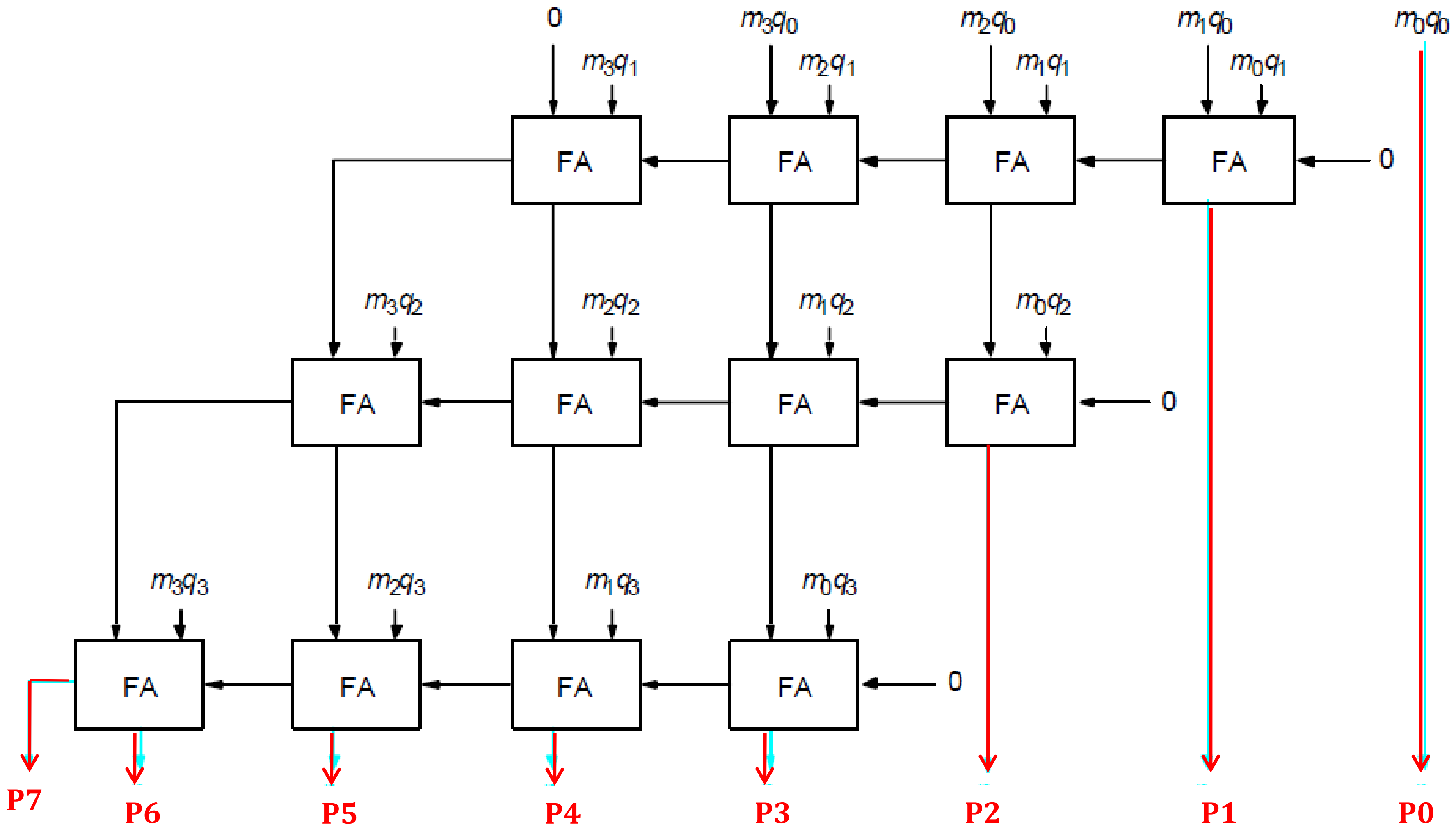
```
                0  1  1  0  1
                0 -1 +1 -1  0
               ---------------
 0  0  0  0  0  0  0  0  0  0
 1  1  1  1  1  0  0  1  1
 0  0  0  0  1  1  0  1
 1  1  1  0  0  1  1
 0  0  0  0  0  0
-----------------------------
 1  1  1  0  1  1  0  0  1  0   (- 78)
```

```
                0  1  1  0  1
                0    -1    -2
               ---------------
 1  1  1  1  1  0  0  1  1  0
 1  1  1  1  0  0  1  1
 0  0  0  0  0  0
-----------------------------
 1  1  1  0  1  1  0  0  1  0
```

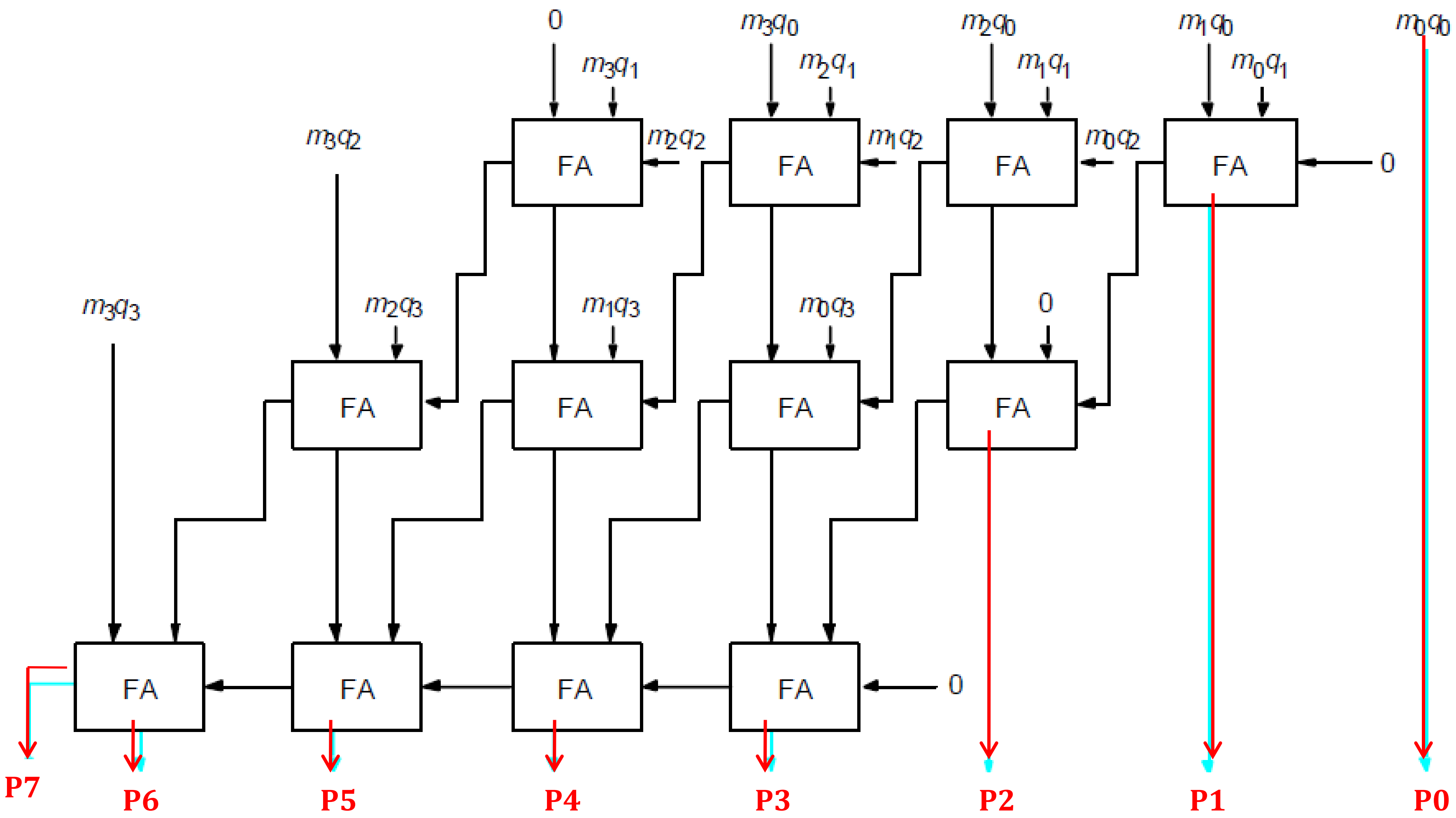
Figure 6.15. Multiplication requiring only *n*/2 summands.

Carry-Save Addition of Summands

- CSA speeds up the addition process.



Carry-Save Addition of Summands (Cont.,)



Carry-Save Addition of Summands (Cont.,)

- Consider the addition of many summands, we can:

➢ Group the summands in threes and perform carry-save addition on each of these groups in parallel to generate a set of S and C vectors in one full-adder delay

➢ Group all of the S and C vectors into threes, and perform carry-save addition on them, generating a further set of S and C vectors in one more full-adder delay

➢ Continue with this process until there are only two vectors remaining

➢ They can be added in a RCA or CLA to produce the desired product

Carry-Save Addition of Summands

						1	0	1	1	0	1	(45)	M
					x	1	1	1	1	1	1	(63)	Q
						1	0	1	1	0	1	A	
					1	0	1	1	0	1		B	
				1	0	1	1	0	1			C	
			1	0	1	1	0	1				D	
		1	0	1	1	0	1					E	
	1	0	1	1	0	1						F	
1	0	1	1	0	0	0	1	0	0	1	1	(2,835)	Product

Figure 6.17. A multiplication example used to illustrate carry-save addition as shown in Figure 6.18.

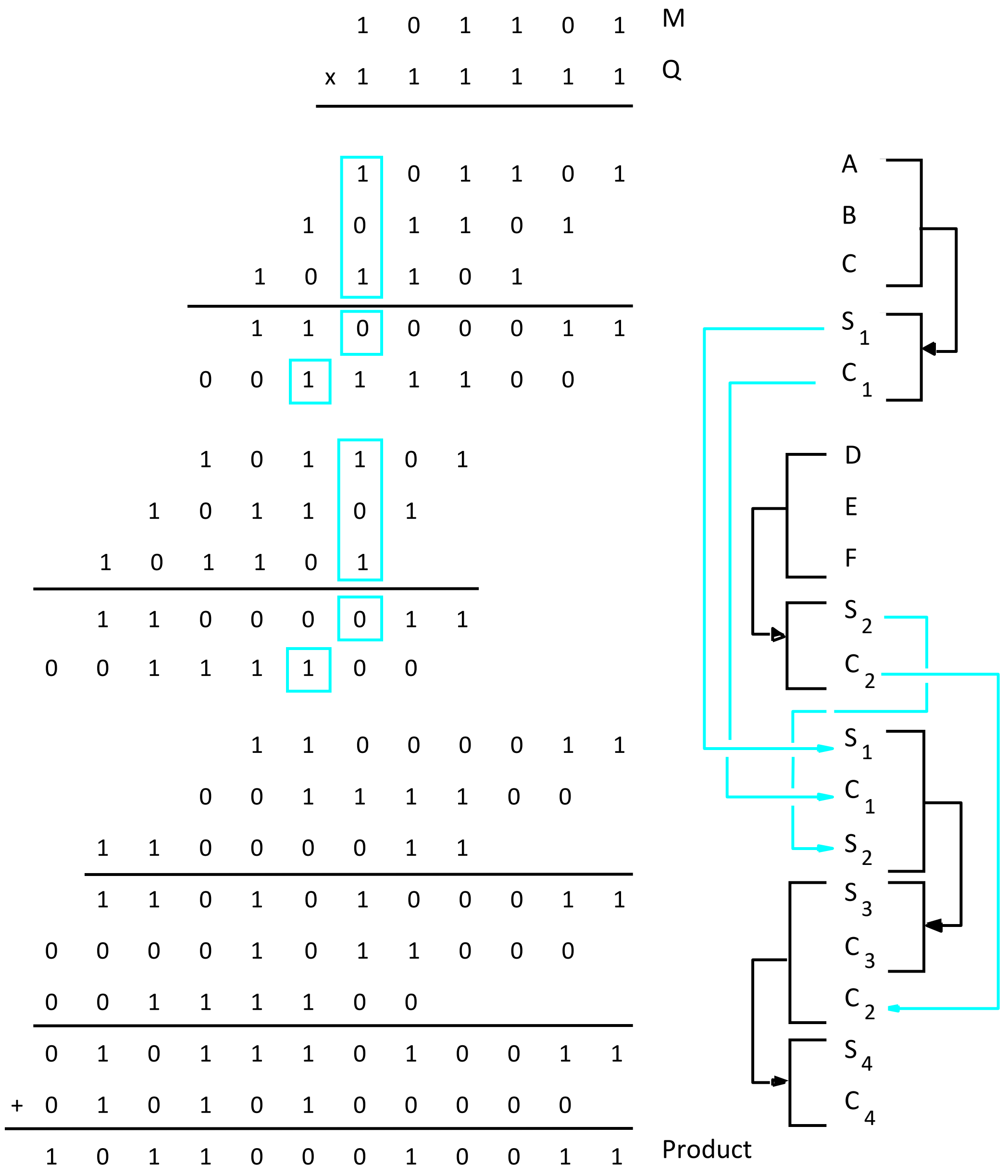


Figure 6.18. The multiplication example from Figure 6.17 performed using carry-save addition.

Integer Division

Manual Division

```
        21                    10101
    ______             ______________
13 ) 274        1101 ) 100010010
     26                 1101
     ___                _____
      14                 10000
      13                  1101
      ___                 _____
        1                   1110
                            1101
                            ____
                               1
```

Longhand division examples.

Longhand Division Steps

- Position the divisor appropriately with respect to the dividend and performs a subtraction.
- If the remainder is zero or positive, a quotient bit of 1 is determined, the remainder is extended by another bit of the dividend, the divisor is repositioned, and another subtraction is performed.
- If the remainder is negative, a quotient bit of 0 is determined, the dividend is restored by adding back the divisor, and the divisor is repositioned for another subtraction.

Circuit Arrangement

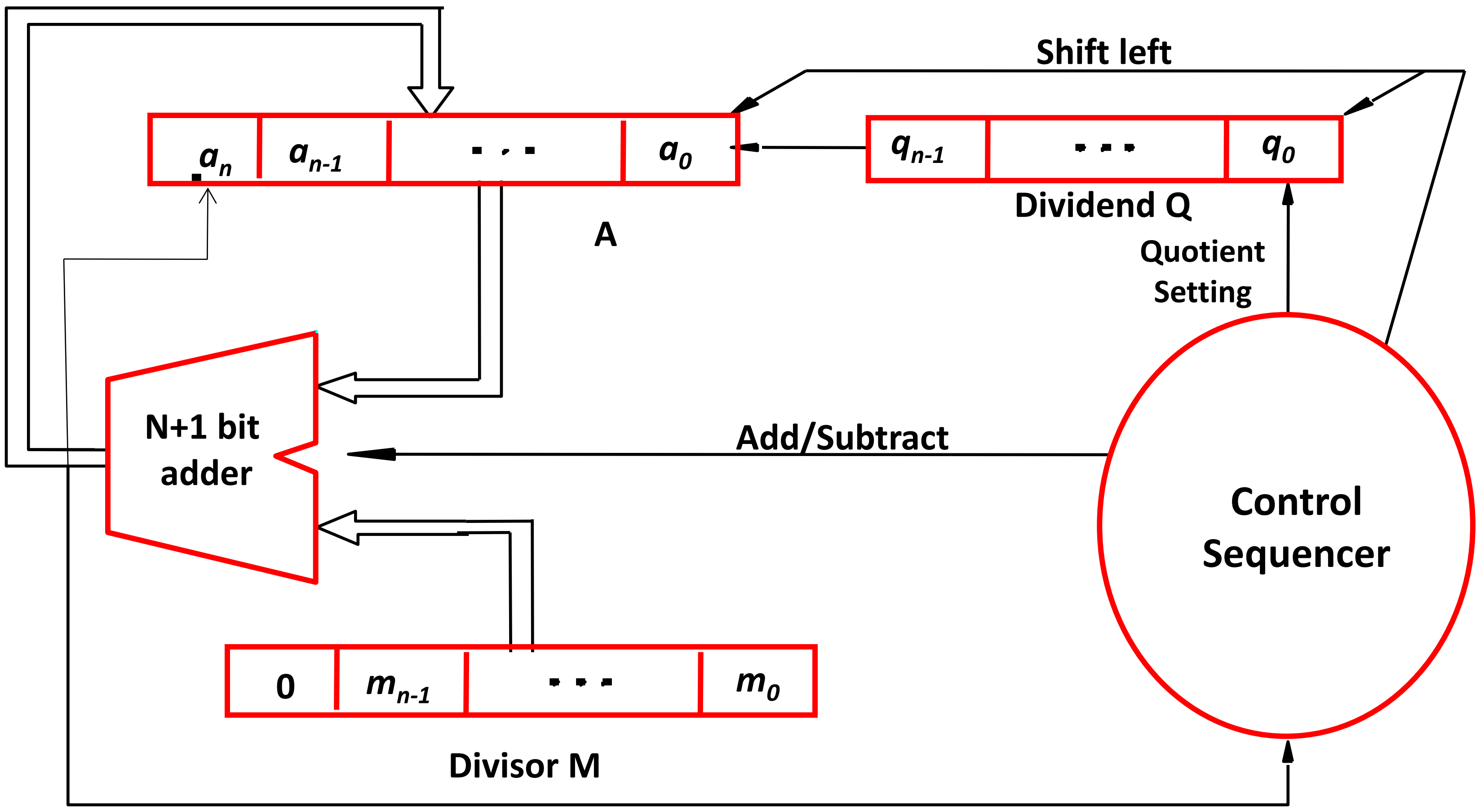


Figure 6.21. Circuit arrangement for binary division.

Restoring Division

- Shift A and Q left one binary position
- Subtract M from A, and place the answer back in A
- If the sign of A is 1, set q_0 to 0 and add M back to A (restore A); otherwise, set q_0 to 1
- Repeat these steps *n* times

Examples

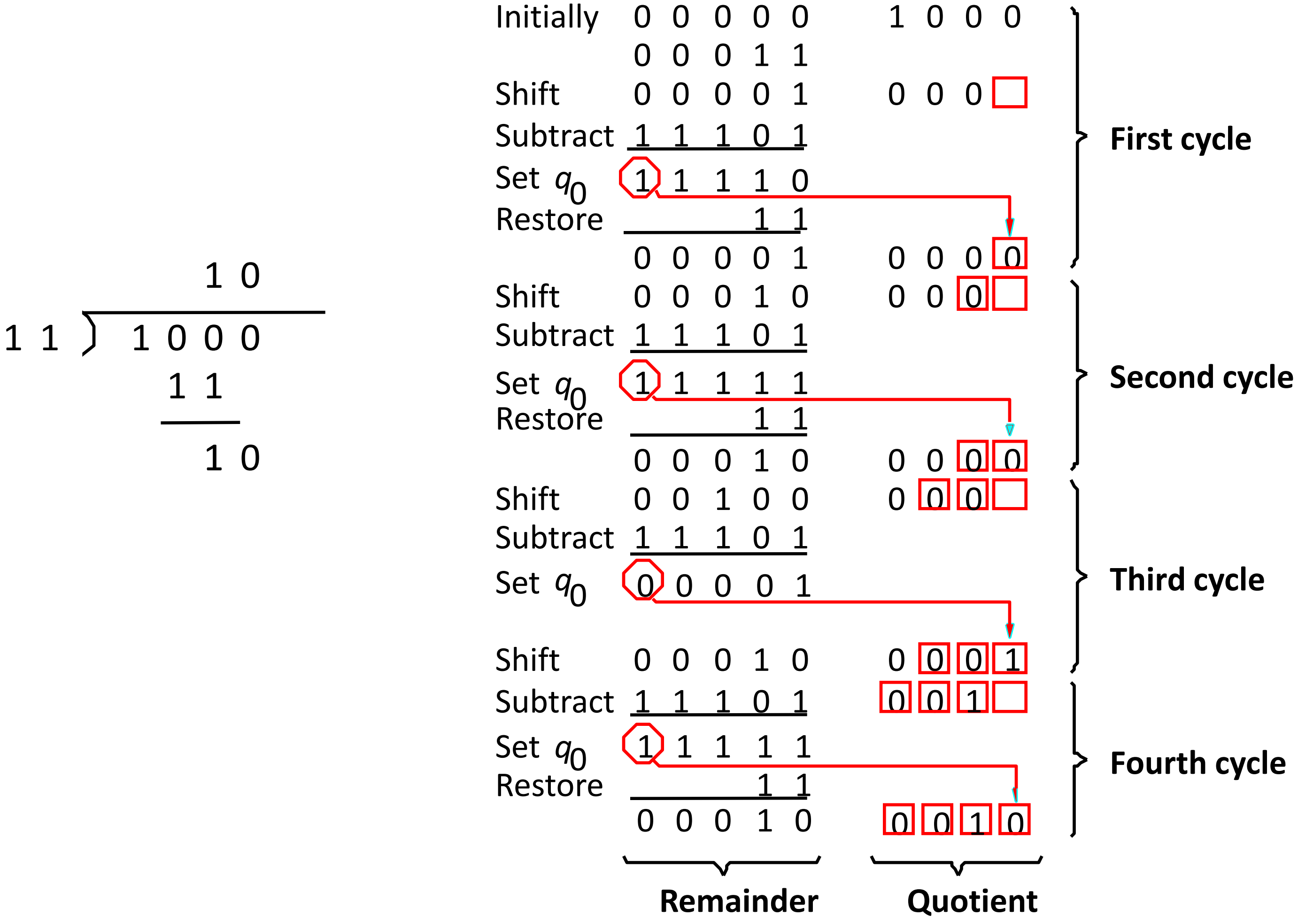


Figure 6.22. A restoring-division example.

Nonrestoring Division

- Avoid the need for restoring A after an unsuccessful subtraction.
- Step 1: (Repeat *n* times)

➢ If the sign of A is 0, shift A and Q left one bit position and subtract M from A; otherwise, shift A and Q left and add M to A.

➢ Now, if the sign of A is 0, set q_0 to 1; otherwise, set q_0 to 0.

- Step2: If the sign of A is 1, add M to A

Examples

Initially	0	0	0	0	0	1	0	0	0	First cycle
	0	0	0	1	1					
Shift	0	0	0	0	1	0	0	0	$\boxed{}$	
Subtract	1	1	1	0	1					
Set q_0	1	1	1	1	0	0	0	0	$\boxed{0}$	
Shift	1	1	1	0	0	0	0	$\boxed{0}$	$\boxed{}$	Second cycle
Add	0	0	0	1	1					
Set q_0	1	1	1	1	1	0	0	$\boxed{0}$	$\boxed{0}$	
Shift	1	1	1	1	0	0	$\boxed{0}$	$\boxed{0}$	$\boxed{}$	Third cycle
Add	0	0	0	1	1					
Set q_0	0	0	0	0	1	0	$\boxed{0}$	$\boxed{0}$	$\boxed{1}$	
Shift	0	0	0	1	0	$\boxed{0}$	$\boxed{0}$	$\boxed{1}$	$\boxed{}$	Fourth cycle
Subtract	1	1	1	0	1					
Set q_0	1	1	1	1	1	$\boxed{0}$	$\boxed{0}$	$\boxed{1}$	$\boxed{0}$	
						Quotient				

Restore remainder

	1	1	1	1	1
	0	0	0	1	1
Add	0	0	0	1	0
	Remainder				

A nonrestoring-division example.